

playground

Picking a Visualization Angle

There are a number of things we can choose to communicate:

1. Monthly/seasonal patterns of frog observations
2. Amount of frogs in each species observed
3. The range (distance) of frog observations
4. Regions where frogs were observed
5. When observations were made
6. How rare/abundant each frog is

And given that the data spans only 2023, we can use:

1. Monthly or seasonal patterns
2. Range of observations
3. Amount of observations
4. Regions of observations
5. When are the most observations made
6. Rarity

I would like to show rarities, and monthly or seasonal patterns on a map (range) for frog species

Exploration & Wrangling

Supplying Common Name for all Observations

I have found a solution in the chunk above, where for frogs which have been recorded only at the species level, I combine their names in the common name column. So we have things like “Alpine Tree Frog / Whistling Tree Frog”.

Splitting Event Date into Month and Year

Correcting Column Data Types

I want to create a new column called “Species”, taken from the last word in each observation’s Scientific Name. OR Factor the commonNames column

In addition to correcting the dtypes of other columns

Exploratory Visualizations

Since we are visualizing monthly/seasonal variations, rarity, and range, I will visualize:

1. The monthly variations using the commonNames and a faceted bar chart. I would need to group by month (extracted from `eventDate`)
2. The rarity using all the the first and last 5 commonNames, and a bar chart
3. The range using by calculating the minimum and maximum longitudes and latitudes to construct a square whose area I can calculate.

Visualizing Top and Bottom 5 across months

NEW PLAN

To ensure that I don’t get stuck on irrelevant things, I am going to pick one of the questions on the readme page, and try to visualize it well. I am going with: Do different frog species have distinct calling seasons?

For this, I will visualize all the frog observations in 12 australian maps (representing the months of the recording year), and point out the most prominent frog in each month.

Additionally, I will highlight the overall most abundant and rare frog in the entire frog list.

How do we plot maps in R?

Good question.

I have found that ggplot is able to do so!

From the tutorial I saw:

Now, our dataset already contains lat and longs so we can plot right away:

I’ve definitely seen worse plots.

Now, let's attempt to facet it by month:

That seems to have worked, but it doesn't look great.

Going back to our visualization questions,

NEW NEW PLAN

I am going to convert the Genuses of all the frogs into its own factor column. Then, using the one map, I will show the distributions of the Genuses across Australia. I will also add pictures of the most prominent frogs in each region.

Now, I am going to visualize (using bar plots and the map), the subfamilies, tribes, and genres of all the frogs in the dataset:

Seeing the plots, I think the Genus level plots are too busy. I will focus on tribe and subfamilies. I just need to find color palettes that don't look awful on the plot.

I am going to try to create separate map heatmap (or choropleth) plots of each tribe or subfamily.

I think I like the tribe plot.

Since we also asked "Do different frog species have distinct calling seasons?", I would like to show the temporal distribution of observations within each tribe across the year, while making sure that the plot is not cluttered.

I can have 6 columns, and within each one, show the scattering of frogs at the four seasons of the year.

There will be 24 maps total. Do I want that?

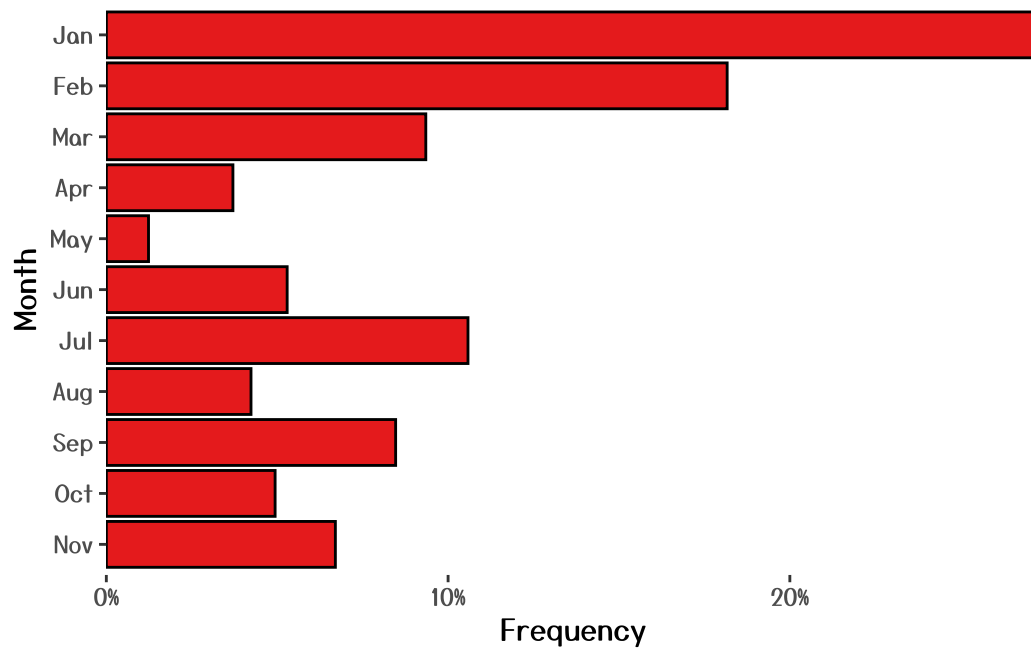
If it is cluttered, I can instead add a single histogram to each map plot to show the frequency of observations throughout the year.

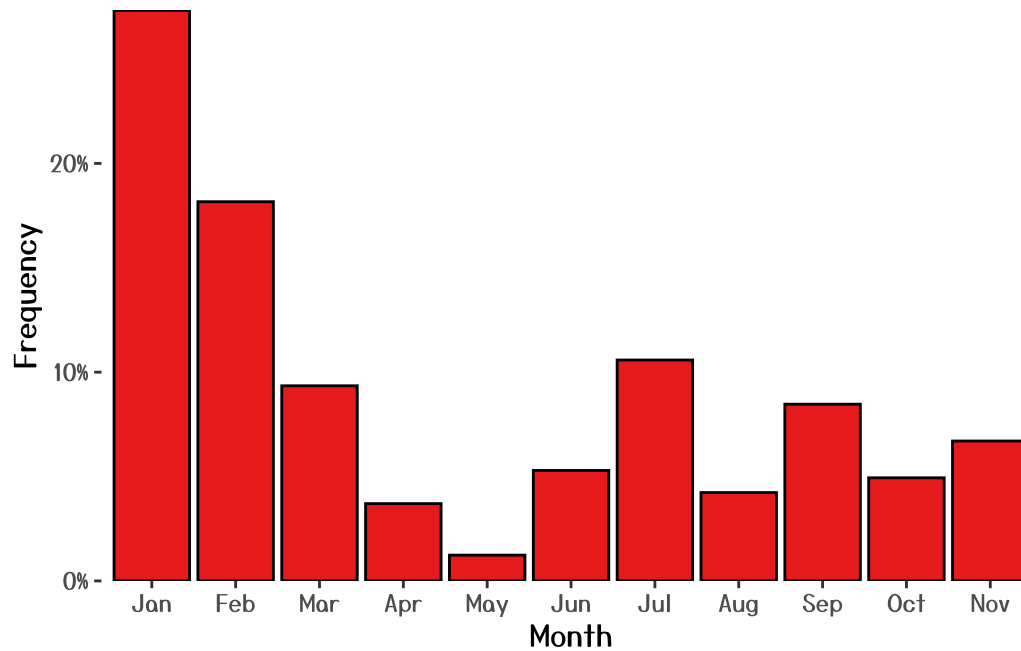
So the multiple maps is too busy. Let's see the histograms alongside the map plots:

Hello Again

I am not going to fuss around with patchwork anymore until I can understand it better. Henceforth, I am going to export the individual plots as svg, and do the finishing touches in Inkscape.

Testing with just two





NOW, HOW ABOUT A STREAM GRAPH?

This isn't bad. Now, I want to:

1. crimp the edges of the plot
2. remove the y-axis ticks
3. match the colours on the stream graphs and the map plots
4. set the global font

